

FMCG AI Assistant

Empowering Business Decisions through Conversational AI

1. Executive Summary

The Core Problem:

Business users frequently request ad-hoc analysis for promotions, inventory, and sales. These requests rely heavily on manual dashboarding by data analysts, leading to delayed decision-making and analytical bottlenecks.

The Proposed Solution:

An AI-powered conversational workspace. Business users ask plain-English questions, and the AI securely translates them into SQL to extract instant insights from the FMCG data warehouse.

2. The 'Wow' Factor / Demo

Imagine a marketing manager asking:

"Show me the revenue impact of the Price Cut on Spark Lemon in the North region."

Within seconds, the Streamlit interface returns the exact revenue figures and unit volumes, without writing a single line of SQL or opening a complicated BI tool.

(Refer to the Streamlit App Demo for the live interface)

3. Data Foundation

Good AI requires excellent data structures. Our foundation is built on an FMCG Star Schema:

- Product Master: Dimension table for categories, sub-categories, and pricing.
- Store Master: Dimension table for geographic regions and store formats.
- Sales & Promotions: Fact table capturing weekly revenue and promotional lifts.
- Inventory: Fact table capturing stock movement and stockout flags.

This structured approach enables the LLM to write highly accurate, deterministic SQL queries.

4. System Architecture

Our architecture ensures a seamless flow from intent to insight:

1. User Interface: A modern Streamlit chat interface.
2. Orchestration: LangChain SQL Agent processes the request.
3. Intelligence Engine: Google Gemini or OpenAI parses the semantic intent.
4. Execution: Validated SQL runs securely against the local SQLite database.
5. Response: Data is synthesized back into natural language for the user.

5. Mitigating Risks

Enterprise AI demands strict guardrails to prevent hallucinations and data leaks:

- Read-Only Execution: The agent is bounded by database credentials that can only SELECT.
- Schema Validation: Queries are validated against the actual schema before execution.
- Context Injection: The prompt enforces strict domain rules (e.g., matching text with LIKE to handle minor misspellings, using specific columns for 'revenue').

6. Future Roadmap (Version 2.0)

This prototype lays the groundwork for enterprise-wide scaling:

- Cloud Integration: Migrate from local SQLite to Snowflake/BigQuery.
- Governance: Implement Row-Level Security so users only query their authorized regions.
- Dynamic Visualizations: Allow the AI to generate Python-based charts on the fly.
- Predictive Capabilities: Forecast stockouts before they happen based on current trajectories.